

SHAHZEB RAHMAN

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PROFESSIONAL SUMMARY

Results-driven **Chemical Engineer & AMPP Certified Basic Coating Inspector (Cert No. 119143)** with extensive expertise in surface preparation, industrial chemical cleaning, passivation, and corrosion mitigation across major oil, gas, and petrochemical facilities. Proven track record in executing high-risk pre-commissioning and maintenance operations while strictly maintaining compliance with international industrial standards (AMPP, NACE, ASTM, ISO). Expert in P&ID analysis, risk assessments (HAZOP, PSSR), and safety protocol management to seamlessly drive operational efficiency and minimize plant asset downtime.

CERTIFICATIONS

AMPP Basic Coatings Inspector – Association for Materials Protection and Performance	Cert No: 119143
ISO 14001:2015 (EMS) Lead Auditor	IOSH Managing Safely
OSHA Safety Standards Certification	Fire Safety & First Aid Training

WORK HISTORY

Chemical Cleaning & Coating Inspector / Supervisor <i>Rawabi Industrial Services (RISAL)</i>	12/2023 – Present <i>Jubail, Saudi Arabia</i>
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- **AMPP Compliance:** Oversee industrial surface preparation and coating application processes, checking ambient conditions (dew point, relative humidity) and dry film thickness (DFT) to guarantee complete compliance with AMPP, NACE, and client specifications.
- **Coating Inspection:** Conduct rigid visual and instrumental inspections of protective coatings on critical industrial assets, identifying defects like pinholes, runs, and sagging to initiate swift remediation.
- **Process Management:** Direct major pre-commissioning and maintenance operations involving specialized chemical cleaning, metal pickling, passivation, and neutralization to safely optimize asset lifetimes.
- **Technical Coordination:** Execute complex site surveys to draft tailored chemical engineering and coating procedures featuring corrosion inhibitors, surfactants, and specific blends tailored to plant parameters.
- **System Isolations:** Map battery limits, determine high-pressure injection/bleeding spots, and establish fluid return networks to streamline operational flow control and safe isolation.
- **Safety Controls:** Review Chemical Hazard Bulletins and Material Safety Data Sheets (MSDS), administering Pre-Startup Safety Reviews (PSSR) and issuing hot work or line-breaking permits for hazardous zones.

Chemical & Pre-Commissioning Engineer <i>Descon Engineering Limited</i>	01/2022 – 11/2023 <i>Karachi, Pakistan</i>
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- **SOP Formulation:** Developed and executed rigorous Standard Operating Procedures (SOPs) for high-pressure systems, introducing precise chemical concentrations and flow loops using diaphragm pumps.
- **Quality Inspection:** Supervised Pre-Mature Inspections (PMI), post-cleaning residue analyses, and pipeline leak testing to verify structural matrix integrity prior to system handovers.
- **Corrosion Control:** Applied specialized chemical corrosion inhibitors to internally safeguard carbon/stainless steel pipelines and high-temperature vessels against rapid oxidation post-decontamination.
- **Risk Mitigation:** Enforced comprehensive Lockout/Tagout (LOTO) structures and interpreted schematic P&IDs, loop diagrams, and NFPA hazard templates to isolate operational energy safely.
- **Compliance & Logistics:** Managed bulk hazardous material stocks and directed effluent handling protocols to preserve full compliance with environmental frameworks (EPA, ISO 14001).

Chemical Engineering Intern	06/2019 – 09/2019
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- **Unit Operations:** Assisted operators in running the Naphtha Hydro Treating (NHT) Unit to reduce organic sulfur and process impurities from targeted chemical streams.
- **Corrosion & Amine Loops:** Monitored chemical rich/lean amine loops for hydrogen sulfide (\$H_2S\$) removal and evaluated systems for foaming, heat exchanger fouling, and contactor corrosion.
- **DCS Monitoring:** Tracked pressure drops, catalyst activity variables, and temperature splits via the Distributed Control System (DCS) to forecast operational bottlenecks.

TECHNICAL CORE SKILLS

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| • Coating Inspection (DFT, Profile) | • Chemical Passivation & Pickling | • Lockout/Tagout & Work Permits |
| • Surface Preparation Standards | • P&ID & Schematic Loop Analysis | • Environmental Compliance (ISO |
| • Corrosion Mitigation | • HAZOP & PSSR Review 14001) | • Project Controls & Scheduling |

EDUCATION

Bachelor of Science in Chemical Engineering (Grade: A)	Graduation: 12/2021
University of Engineering and Technology (UET) Peshawar	Peshawar, Pakistan
Final Year Project: Bagasse & Rice Husk Matrix as Cement Substitutes (Reduced \$CO_2\$ by 10%)	

PERSONAL DETAILS

Visa Status: Transferable Work Visa (KSA) | **Nationality:** Pakistani | **Languages:** English (Advanced), Urdu (Proficient), Pushto (Proficient)